

SPECIFICATION

产品规格书

This module uses ROHS material

模块用环保材料

Customer Nam 客户名称	
Project Name 成品机型	XY347001C18
Date 日期	2024.7.1

Customer' s Acceptance 客户承认:

Comment 承认意见	Approved by 承认人

Written by 撰写	Approved by 批准

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湖南旭阳显示科技有限公司

Hu Nan xuyang photoelectric Group Co., Ltd

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1. GENERAL INFORMATION AND Electro Optical Characteristics

主要特征描述和光电参数

1.1 GENERAL INFORMATION主要特征描述

Item 项目	Contents 内容	Unit 单位
LCD Type 液晶显示类型	TFT/TRANSMISSIVE	---
Viewing Direction 视角方向	ALL	O’Clock
Outline Dimensions (W × H×T) 外形尺寸 （宽 x 高 x 厚）	54.5(W) ×83(H)×2.1(T)	mm
Viewing area 可视区域	49.96(W) x74.44(H)	mm
Active area 有效区域 (宽 × 高)	48.96(W) x73.44(H)	mm
Number of Dots 点阵	212RGB x 318 Dots	---
Pixel pitch (W × H) 像素点尺寸	0.051×0.153	mm
Driver IC 驱动 IC	ST7789P3	---
Interface Type 接口类型	MCU8BIT	---
Input voltage 输入电压	2.8V	—
Colors 色彩	265K	
Backlight Type 背光类型	LED	---

1.2 Electro Optical Characteristics光电参数

Item of electro-optical characteristics 项目	Symbol 符号	Condition 条件	Min 最小值	Typ 典型值	Max 最大值	Unit 单位	Remark 注 释
Surface Luminance 表面亮度	Lv	$\theta = 0^\circ$ $\psi = 0^\circ$	180	250	-	Cd/m ²	
Luminance uniformity 均匀度	δ WHITE	I _f =20mA /LED	80	-	-	%	

Item	Symbol	Condition	Values			Unit	Notes
			Min.	Typ.	Max.		
CT1 Transmittance	TR%	/	/	/	/	%	
CT2 Transmittance	TR%	/	4.74%	5.27%	/	%	@C light +DCI 光阻+ 非导电 POL +w/o APF+ w/o haze
Response time	Tr+ Tf	/	/		30	ms	/
Contrast ratio	CR	/	/	/	/	/	/
Viewing angle (CR≥ 10)	θL	Φ=180°	70	80	/	/	/
	θR	Φ=0°					
	θT	Φ=90°					
	θB	Φ=270°					
Color chromaticity (CIE1931)	Wx	/	-0.02	0.321	+0.02	/	@C-light
	Wy	/		0.354		/	
	Rx	/	-0.015	0.668	+0.015	/	
	Ry	/		0.327		/	
	Gx	/		0.266		/	
	Gy	/		0.631		/	
	Bx	/		0.134		/	
	By	/		0.098		/	
CF NTSC	/	/	/	85%	/	/	/
Crosstalk	/	/	/	/	2	%	
Flicker	/	/	/	-30	/	dB	

Note 1: Ambient temperature = 25°C.

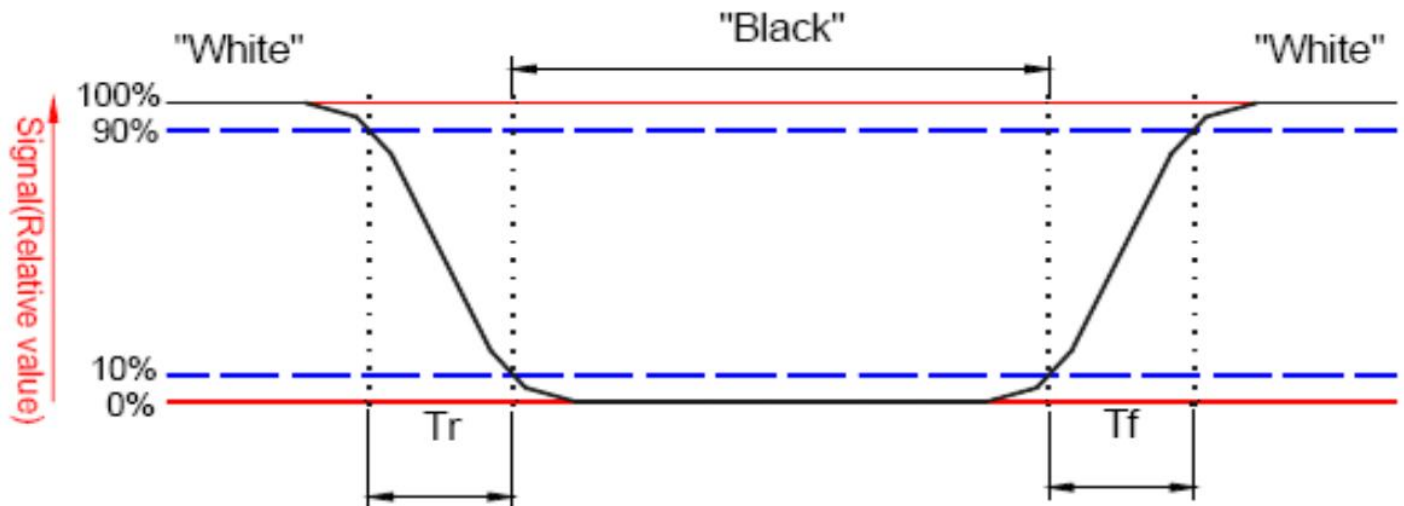
Note 2: To be measured with a viewing cone of 2° by Topcon luminance meter BM-5A.

Note 3: To be measured with Otsuta chromaticity meter LCF-2100M, CF only measure under C light simulation.

Note 4: CTC shipping status is cell without polarizer. Transmittance of Specification is cell with polarizer. The tolerance of Transmittance is ±10%.

Note 5: Definition of response time:

The output signals of TRD-100 are measured when the input signals are changed to "White" (falling time) and from "White" to "Black" (rising time), respectively. The interval is between the 10% and 90% of amplitudes. Refer to figure as below.



Note 6: Definition of contrast ratio:

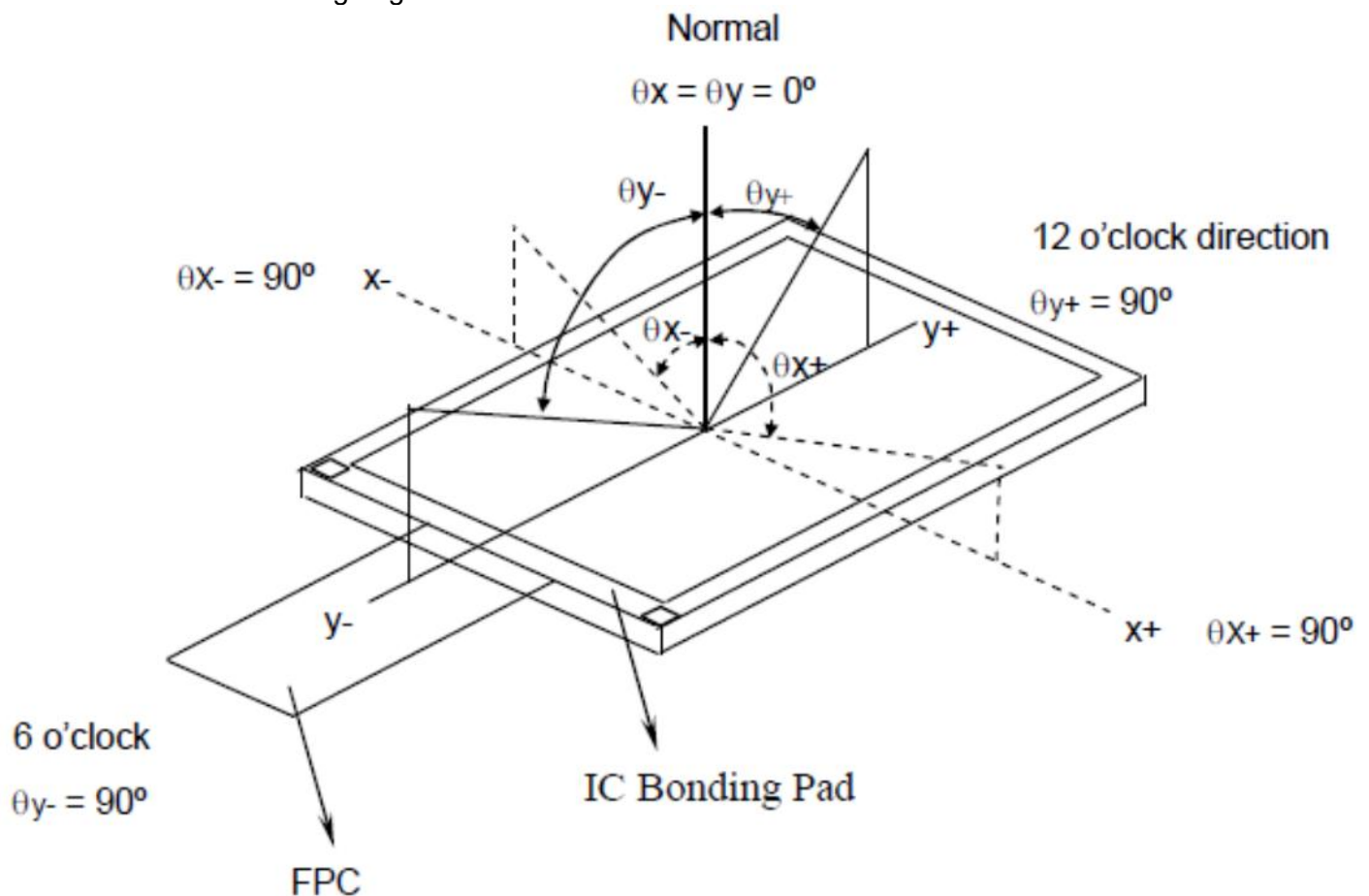
Contrast ratio is calculated by the following formula.

Brightness on the "white" state

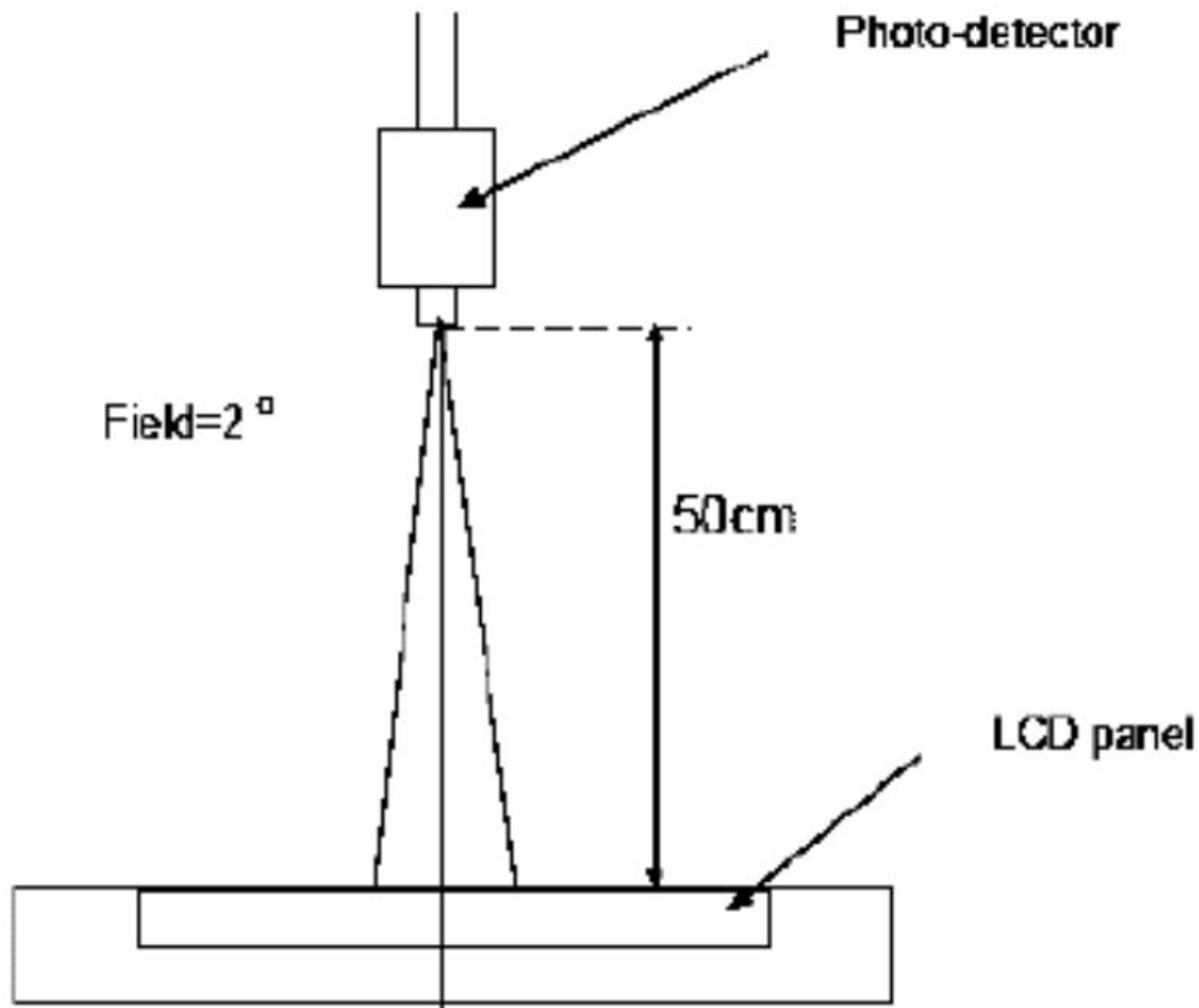
Contrast ratio (CR)=

Brightness on the "black" state

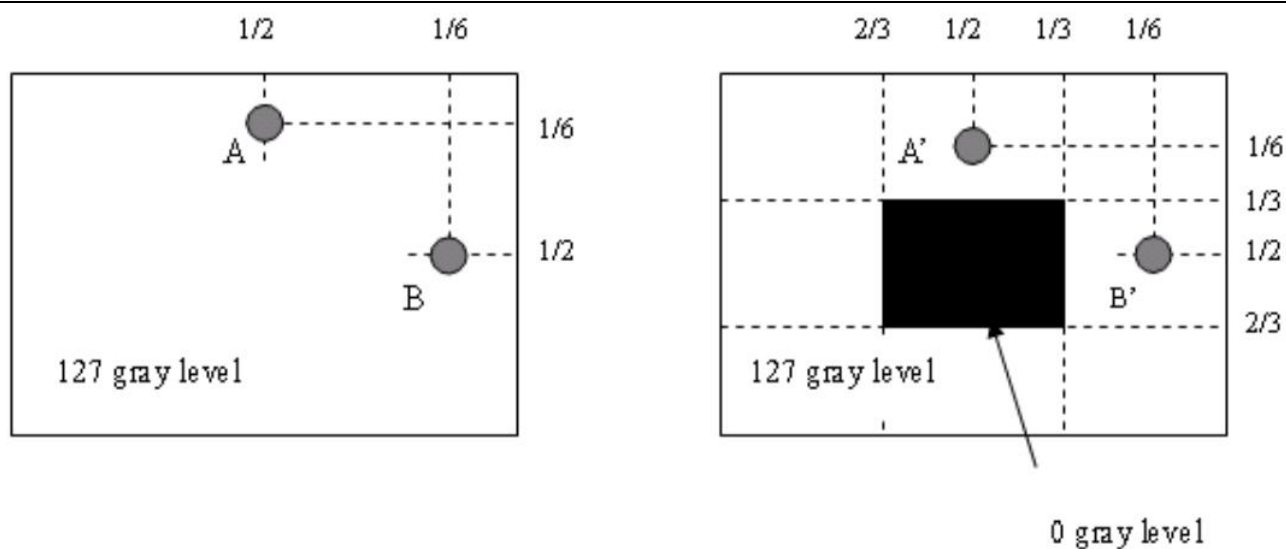
Note 7: Definition of viewing angle



Note 8: Optical characteristic measurement setup.

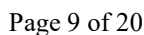


Note 9:



$|LA - LA'| / LA \times 100\% = 2\% \text{ max.}$, LA and LA' are brightness at location A and A'.

$|LB - LB'| / LB \times 100\% = 2\% \text{ max.}$, LB and LB' are brightness at location B and B'.



3. ELECTRICAL CHARACTERISTICS 模块电气特性

3.1 DC CHARACTERISTICS 直流特性

Item 项目	Symbol 符号	Min 最小值	Typ 典型值	Max 最大值	Unit 单位
Supply voltage for logic 逻辑电压	VDDIO	1.65	1.8/2.8	3.3	V
Supply Voltage for analog 模拟电压	VCI	2.5	2.8	3.3	
Input Current 输入电流	Idd		TBD	TBD	mA
Input voltage 'H' level 输入高电平	VIH	0.7VDDIO	—	VDDIO	V
Input voltage 'L' level 输入低电平	VIL	VSS		0.3VDD	V
Output voltage 'H' level 输出高电平	VOH	0.8VDDIO		VDDIO	V
Output voltage 'L' level 输出低电平	VOL	VSS		0.2VDDIO	V

3.2 BACKLIGHT CHARACTERISTICS 背光电气特性

Item 项目	Symbol 符号	Min 最小值	Typ 典型值	Max 最大值	Unit 单位	Condition 条件
Forward voltage 正向电压	Vf	3.0	3.2	3.4	V	If=6*20mA
Number of LED LED数量			6		Piece	
Connection mode 连接类型	P		Parallel			
LED life time寿命	-		15000		Hr	Note 1

Note:

(1) The “LED life time” is defined as the module brightness decrease to 50% of original brightness

at IL=120mA. The LED life time could be decreased if operating IL is larger than 120mA.

Backlight circuit diagram shown in below:

4. INTERFACE DESCRIPTION 接口定义描述

Pin No.	Symbol	I/O	Function
1	BL-K		LED Negative
2	BL-A		LED Positive
3	GND		GROUND
4	VDD		Power Supply
5	/RSTB		System Reset Pin
6	RDB		Read enable clock input pin
7	WRB		Write enable clock input pin
8	RS		Register select signal pin. L: Command H: Data
9	/CS		Chip select input pin
10~17	DB7-DB0		Data Bit
18	GND		GROUND

Note: The voltage power of the interface logic pin depend on IOVCC and GND, Such as DBn, IMn and function pins

5. Interface Timing.

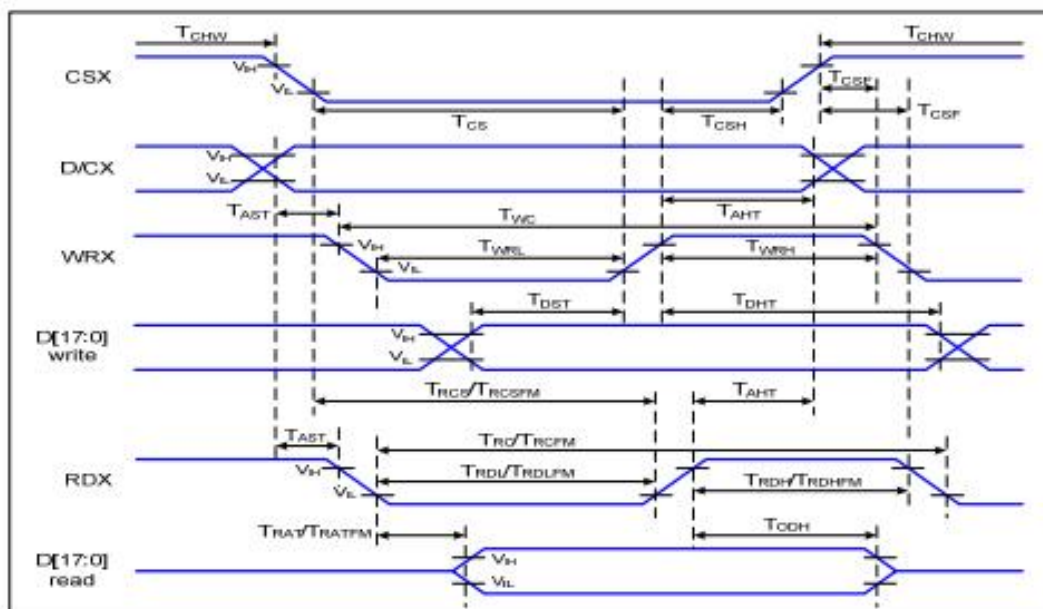
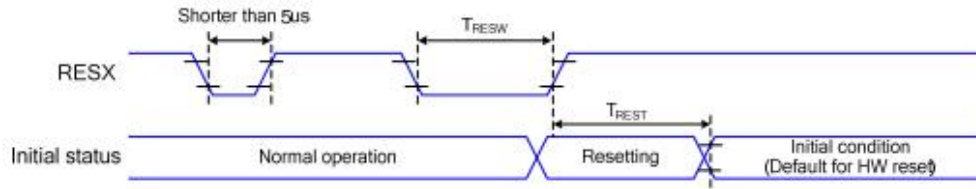


Figure 1 Parallel Interface Timing Characteristics (8080-Series MCU Interface)

VDD=1.65 to 3.3V, VDD=2.4 to 3.3V, AGND=DGND=0V, Ta= -30 to 70 °C

Signal	Symbol	Parameter	Min	Max	Unit	Description
D/CX	T _{AST}	Address setup time	0		ns	-
	T _{AHT}	Address hold time (Write/Read)	10		ns	
CSX	T _{CHW}	Chip select "H" pulse width	0		ns	-
	T _{CS}	Chip select setup time (Write)	15		ns	
	T _{RCS}	Chip select setup time (Read ID)	45		ns	
	T _{RCSFM}	Chip select setup time (Read FM)	355		ns	
	T _{CSF}	Chip select wait time (Write/Read)	10		ns	
	T _{CSH}	Chip select hold time	10		ns	
WRX	T _{WC}	Write cycle	66		ns	-
	T _{WRH}	Control pulse "H" duration	15		ns	
	T _{WRL}	Control pulse "L" duration	15		ns	
RDX (ID)	T _{RC}	Read cycle (ID)	160		ns	When read ID data
	T _{RDH}	Control pulse "H" duration (ID)	90		ns	
	T _{RDL}	Control pulse "L" duration (ID)	45		ns	
RDX (FM)	T _{RCFM}	Read cycle (FM)	450		ns	When read from frame memory
	T _{RDHFM}	Control pulse "H" duration (FM)	90		ns	
	T _{RDLFM}	Control pulse "L" duration (FM)	355		ns	
D[17:0]	T _{DST}	Data setup time	10		ns	For CL=30pF

6. Reset Timing



Related Pins	Symbol	Parameter	MIN	MAX	Unit
RESX	t_{RESW}	Reset Pulse Duration	10	-	us
	t_{REST}	Reset Cancel	-	5	ms
				120	ms

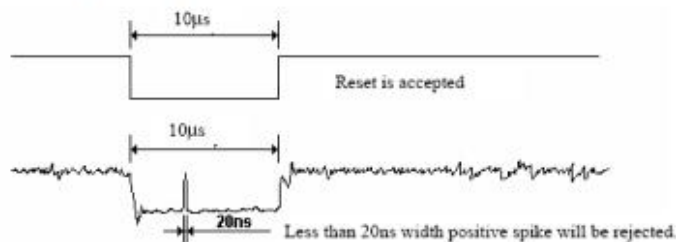
Table 14 Reset Timing

Notes:

1. The reset cancel includes also required time for loading ID bytes, VCOM setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time (t_{RT}) within 5 ms after a rising edge of RESX.
2. Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below:

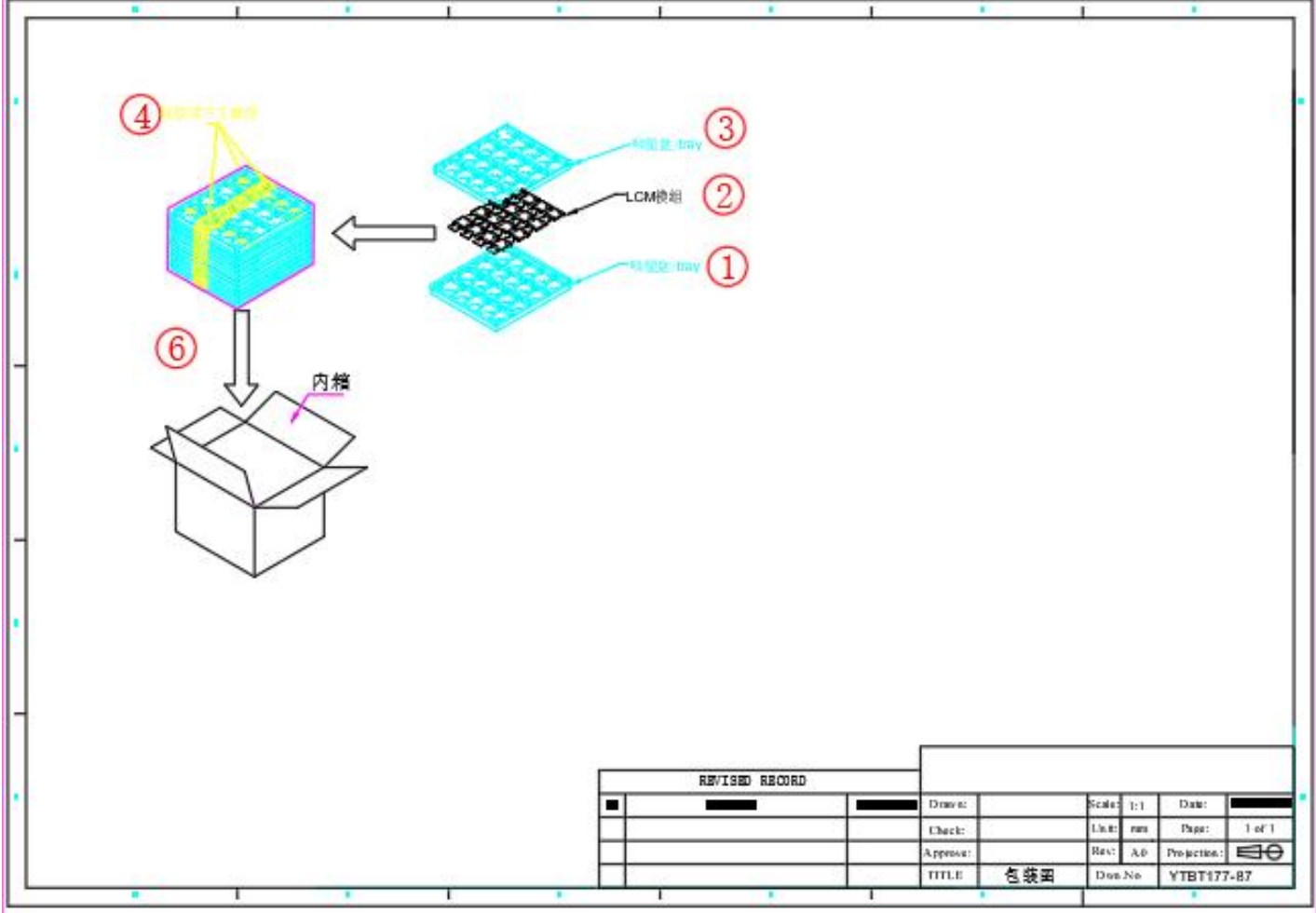
RESX Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 9us	Reset
Between 5us and 9us	Reset Starts

3. During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In -mode.) and then return to Default condition for Hardware Reset.
4. Spike Rejection also applies during a valid reset pulse as shown below:



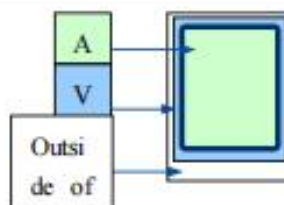
5. When Reset applied during Sleep In Mode.
6. When Reset applied during Sleep Out Mode.
7. It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

7.Packaging details is missing

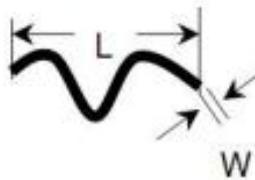


8.INSPECTION CRITERION

显示模块出货检验标准	
一、内容:	
1、抽样方案 (Sampling & Acceptable Quality Level) : 按照抽样方案 GB/T2828.1-2003, 一般检查水平II级, 划分产品可以接受或拒绝的等级如下: 重缺陷: 采用 AQL 0.65 轻缺陷: 采用 AQL 1.0	
2、检查条件: 在 20~40W 日光灯的光照环境下, 被检查样品放在离检查者眼睛 30cm 的位置, 检查者在垂直方向 45 度范围内观察。	
3、检查区域定义(Definition Of Inspection Area): A.A: Active Area V.A: Viewing Area	

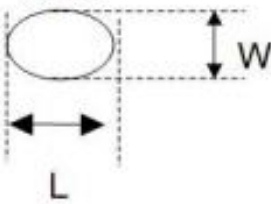
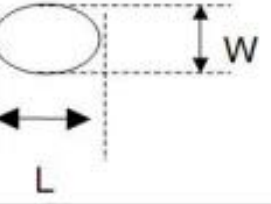


4、外观检查标准(Visual inspection criterion in cosmetic)

玻璃缺陷(Glass defect)					
序号 (NO)	缺陷 (Defect)	缺陷 等级	检验标准 (Criteria)		备注 (Remark)
1	尺寸 (dimension)	轻 (Minor)	按照图纸 (By engineering diagram)		
2	玻璃 (Cracks)	重 (Major)	延伸性破裂 (Extensive crack) 【拒收 Reject】		
LCD V.A.区内的外观缺陷(LCD appearance defect with in V.A.)					
序号 (NO)	缺陷 (Defect)	缺陷 等级	检验标准 (Criteria)		备注 (Remark)
1	毛丝,划痕 (Fiber, scratches)	轻 (Minor)	缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	1.L: Length, W: Width; 2、Disregard if out of V. A. 
			W ≦ 0.03mm	忽略不计 (Disregard)	
			L ≦ 4.0mm, 0.03mm < W ≦ 0.05mm	2	
			L ≦ 4.0mm, 0.05mm < W ≦ 0.08mm	1	
			W > 0.08mm	0	

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2	污点,圆型的 (Dirty Spots)	轻 (Minor)	缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	1. $D = (L+W) / 2$, L: Length, W: Width; 2. Disregard if out of V. A. 
			$D \leq 0.10\text{mm}$	忽略不计 (Disregard)	
			$0.10\text{mm} < D \leq 0.15\text{mm}$	2	
			$0.15\text{mm} < D \leq 0.20\text{mm}$	1	
			$D > 0.20\text{mm}$	0	
3	偏光片凹痕 (Polarizer dent)	轻 (Minor)	缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	1. $D = (L+W) / 2$, L: Length, W: Width; 2. Disregard if out of V. A. 
			$D \leq 0.20\text{mm}$	忽略不计 (Disregard)	
			$0.20\text{mm} < D \leq 0.30\text{mm}$	2	
			$0.30\text{mm} < D \leq 0.50\text{mm}$	1	
			$D > 0.50\text{mm}$	0	

FPC

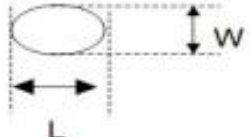
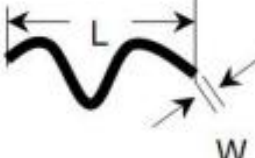
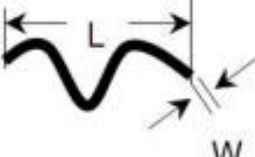
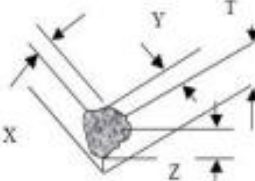
序号 (NO)	缺陷 (Defect)	缺陷 等级	检验标准 (Criteria)	备注 (Remark)
1	铜线剥离 (Copper peeling)	重 (Major)	Not allowed	

硅胶(Silicon)

序号 (NO)	缺陷 (Defect)	缺陷 等级	检验标准 (Criteria)	备注 (Remark)
1	硅胶涂量 (Amount of silicon)	重 (Major)	ITO 暴露 (ITO exposed) 【拒收 Reject】	

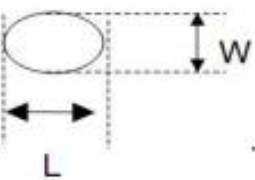
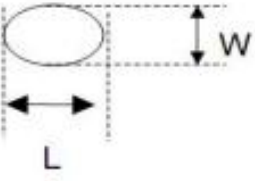
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Touch Panel					
序号 (NO)	缺陷 (Defect)	缺陷 等级	检验标准 (Criteria)		备注 (Remark)
1	污点,圆型 (Dirty Spots, Round type)	轻 (Minor)	缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	1、 $D = (L+W) / 2$, L: Length, W: Width; 2、Disregard if out of V. A. 
			$D \leq 0.1\text{mm}$	忽略不计 (Disregard)	
			$0.1\text{mm} < D \leq 0.20\text{mm}$	2	
			$0.20\text{mm} < D \leq 0.25\text{mm}$	1	
			$D > 0.25\text{mm}$	0	
2	划痕 (Scratches)	轻 (Minor)	缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	1、L:Length, W: Width; 2、Disregard if out of V. A. 
			$L \leq 5.0\text{mm}, W \leq 0.03\text{mm}$	忽略不计 (Disregard)	
			$L \leq 4.0\text{mm}, 0.03\text{mm} < W \leq 0.05\text{mm}$	2	
			$L \leq 3.0\text{mm}, 0.05\text{mm} < W \leq 0.08\text{mm}$	1	
			$L > 2.0\text{mm}, W > 0.08\text{mm}$	0	
3	线缺陷 (Line defect)	轻 (Minor)	缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	1、L:Length, W:Width; 2、Disregard if out of V. A. 
			$L \leq 5.0\text{mm}, W \leq 0.02\text{mm}$	忽略不计 (Disregard)	
			$L \leq 4.0\text{mm}, 0.02\text{mm} < W \leq 0.04\text{mm}$	2	
			$L \leq 3.0\text{mm}, 0.04\text{mm} < W \leq 0.06\text{mm}$	1	
			$L > 3.0\text{mm}, W > 0.06\text{mm}$	0	
4	角破碎 (Corner chip)	轻 (Minor)	$X \leq 3.0\text{mm}, Y \leq 3.0\text{mm}, Z \leq T$		

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			缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	
4	TFT 点缺陷 (TFT Dot defect)	重 (Major)	缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	1、Disregard if out of A. A. 
			亮点 (Bright dot)	1	
			暗点 (Dark dot)	2	
5	显示不均 (Mura)	轻 (Minor)	参照极限样板 (Bye limited sample)		
6	背光污点,圆型 (BL Dirty Spots, Round type)	轻 (Minor)	缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	1、 $D \leq (L+W)/2$, L: Length, W: Width; 2、Disregard if out of V. A.
			$D \leq 0.1\text{mm}$	忽略不计 (Disregard)	
			$0.1\text{mm} < D \leq 0.15\text{mm}$	2	
			$0.15\text{mm} < D \leq 0.2\text{mm}$	1	
			$D > 0.2\text{mm}$	0	
7	划痕、线缺陷 (Scratches、Line defect)	轻 (Minor)	缺陷规格 (Defect Spec.)	允许数量 (Permissible Q'ty)	1.L:Length, W: Width; 2.Disregard if out of V. A. 
			$L \leq 2.0\text{mm}, W \leq 0.02\text{mm}$	忽略不计 (Disregard)	
			$L \leq 3.0\text{mm}, 0.02\text{mm} < W \leq 0.03\text{mm}$	2	
			$L \geq 2.0\text{mm}, 0.03\text{mm} < W \leq 0.05\text{mm}$	1	
			$L \geq 2.0\text{mm}, W > 0.05\text{mm}$	0	

二、其它 (Others)

1. 本标准未尽事宜, 应由双方协商处理。

Issues that are not defined in this document shall be discussed and agreed with both parties. (Customer and supplier)

2. 除非另有书面约定, 否则此标准应适用于自双方签字之日起生效。

Unless otherwise agreed upon in writing, the criteria shall be applied to both parties. (Customer and supplier)

9. Reliability Test Condition and Method

No. 序号	Test Item 试验项目	Test condition 试验条件	Inspection after test 判断标准
1	High Temperature Storage 高温存放	80℃±2℃ 48H	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 试验结束后,已测试的 LCD 样品必须在室内正常温湿度环境下放置 2~4 个小时以上才能进行功能和外观检查, 样品不允许有以下缺陷: 1.Air bubble in the LCD; 模块中有气泡; 2.Sealleak; 封口松脱; 3.Non-display; 不显示; 4.missing segments; 漏笔 5.Glass crack; 玻璃破碎; 6.Current Idd is twice higher than initial value. 电流 Idd 大于初时值的 2 倍
2	Low Temperature Storage 低温存放	-30℃±2℃ 48H	
3	High Temperature Operation 高温操作	70℃±2℃ 48H	
4	Low Temperature Operation 低温操作	-20℃±2℃ 48H	
5	High Temperature /Humidity Storage 高温高湿	60℃±2℃ 90%RH 48H	
6	ESD test 静电试验	C:150pf; R:330Ω ; Voltage:8KV; Air discharge(空气放电),10 time Voltage:4KV; Connet discharge(接触放电),10 time	

10. Precautions for Use of LCD Modules

10.1 The display panel is made of glass. Do not subject it to a mechanical shock by dropping it from a high place, etc.

10.2 If the display panel is damaged and the liquid crystal substance inside it leaks out, be sure not to get any in your mouth, if the substance comes into contact with your skin or clothes, promptly wash it off using soap and water.

10.3 Do not apply excessive force to the display surface or the adjoining areas since this may cause the color tone to vary.

10.4 The polarizer covering the display surface of the LCD module is soft and easily scratched. Handle this polarizer carefully.

10.5 When storing the LCD modules, avoid exposure to direct sunlight or to the light of fluorescent lamps.

10.6 The LCD modules should be stored under the storage temperature range. If the LCD modules will be stored for a long time, the recommend condition is:

Temperature : 0℃ ~ 40℃ Relatively humidity: ≤80%

10.7 The LCD modules should be stored in the room without acid, alkali and harmful gas.

10.8 The LCD modules should be no falling and violent shocking during transportation, and also should avoid excessive press, water, damp and sunshine.

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